Research Highlights

Topological-Relativistic-Thermodynamic Learning (TRTL): Continuous Manifold Invariance, Coherent Wave Superposition, and Certified Adversarial Immunity

IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)

- **Physics-Grounded Classification Paradigm:** Unifies General Relativity (Lorentz metric dilatation), Helmholtz wave optics (coherent phase-interference), and non-equilibrium thermodynamics (Ginzburg-Landau phase crystallization) to replace arbitrary global decision boundaries.
- **Circumvention of High-Dimensional Measure Concentration:** Formulates Relativistic Anisotropic Metric Dilatation (RAMD) driven by local Shannon conflict entropy, sustaining an empirical relative contrast floor of 0.6924 at $D = 1,000$ ($5.2\times$ above Euclidean and Lorentzian baselines).
- **Exact Destructive Horizons & Perfect OOD Detection:** Proves closed-form destructive nodal wave horizons yielding identically zero energy support ($E = 0$) on metric voids, achieving a flawless 1.0000 AUROC on far-OOD queries where deep neural networks exhibit pathological 100% overconfidence.
- **Severe Label Noise Shielding (+35.1% over Deep MLP):** Under 40% symmetric label flip noise, deep neural networks collapse by 43.40% (to 54.49%), while TRTL maintains 89.63% accuracy by dynamically expelling corrupted boundary anchors via surging Lorentz factors ($\gamma_i \gg 1$).
- **Continuous Thermodynamic Counterfactual Recourse (TCR):** Formulates gradient-free recourse along conservative vector fields, achieving a mean trajectory curvature of $\kappa = 0.012$ ($475\times$ smoother than DiCE) with a 100% boundary crossing success rate.
- **Rigorous Theoretical Guarantees & Open Reproducibility:** Provides 11 formal mathematical proofs, including certified ℓ_2 adversarial invariance, Lyapunov monotonic dissipation, and $\mathcal{O}(D \log N)$ Ball-Tree spatial scaling, backed by a permanent citable Zenodo archive (DOI: `10.5281/zenodo.23936990`).